SITEPULSE / AUTONOMOUS FIELD POWER UNIT

Field Operation & Maintenance Guide

How to place, start, monitor and service a SitePulse unit — the onboard battery, the mobile application, the fuel system and the engine.

CONTINUOUS	SURGE	STORAGE	ENGINE
2000 W	**4000 W**	**1545 Wh**	**79 cc**

CONTENTS

AT A GLANCE

BATTERY	$LiFePO_4$, 48 V, 1545 Wh
AC OUT	4 × 120 V, pure sine
SOLAR IN	800 W max, MPPT
WALL IN	120 V, 1500 W max
ENGINE	Predator 79 cc, item 69733
FUEL	87+ octane, ≤10% ethanol
OIL	SAE 10W-30, 0.37 qt
AUX TANK	5 gal metal jerry can

01

Placement and site setup

The unit is designed to run unattended, outdoors. Placement determines whether it can reach the network at all.

01 **Set the unit outside with a clear view of the sky.** The Starlink terminal is mounted under the lid and needs unobstructed sky to acquire and hold a link. Under a roof, a tarp, or heavy canopy the unit will run but will not report.

02 **Level, firm ground.** Set on a stable surface with the exhaust side clear of walls, tents, doorways and air intakes. Never run the engine indoors or in an enclosed space.

03 **Check the lid.** The cover is held by four Velcro tabs; lifting from the sides takes minor force. The Starlink can be relocated off the unit if the site demands it — that requires a longer Ethernet run and power cable.

04 **Confirm airflow.** Both louvered sides must stay unobstructed. The cooling fan draws through them during engine operation.

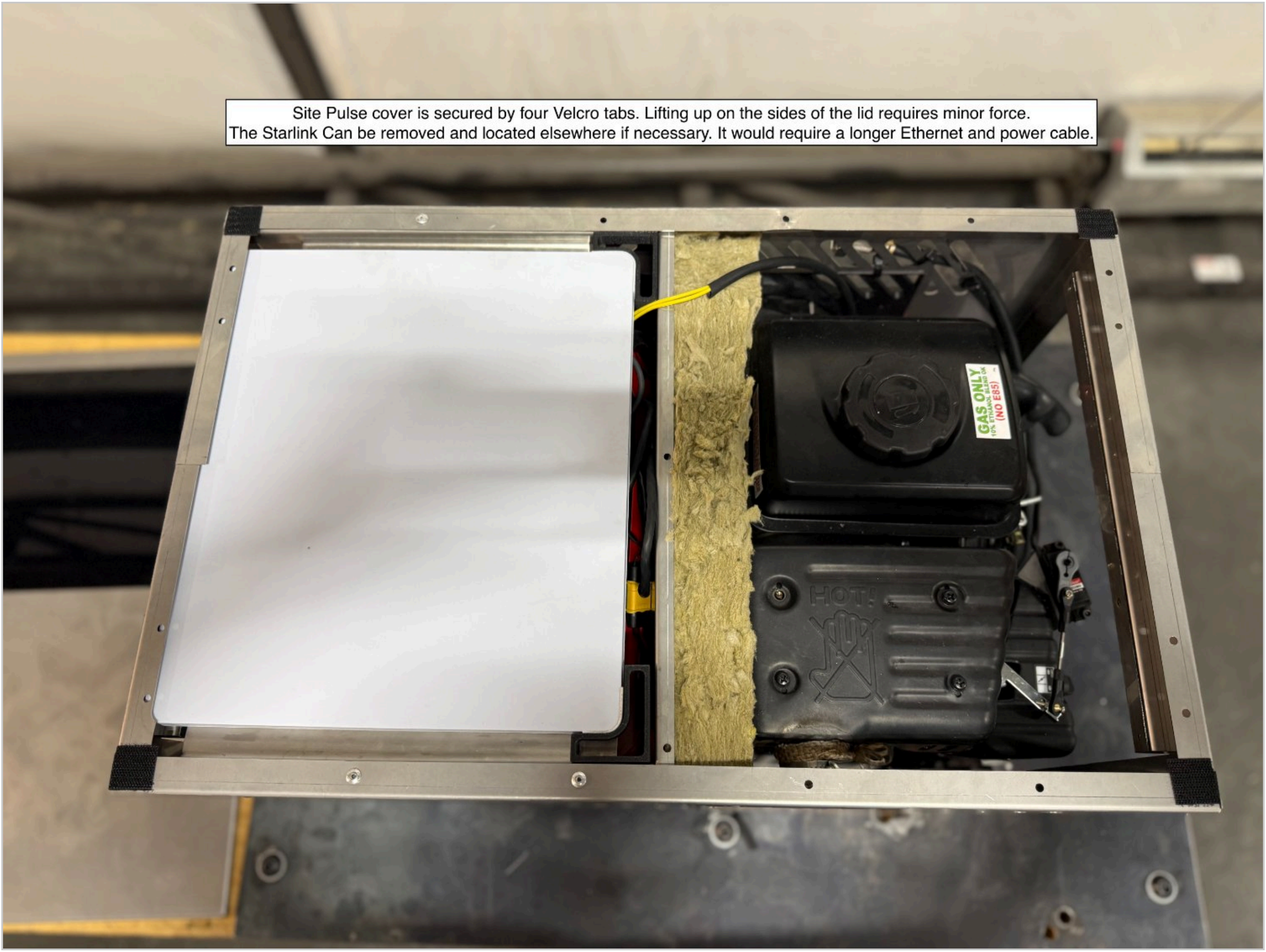

FIG. 01 — LID REMOVED. STARLINK TERMINAL AT LEFT, ENGINE BAY AT RIGHT, INSULATION BETWEEN.

02

Power-up sequence

Two switches must be on. The battery alone will deliver power, but without the onboard computer the engine will not recharge it and the unit cannot be operated remotely.

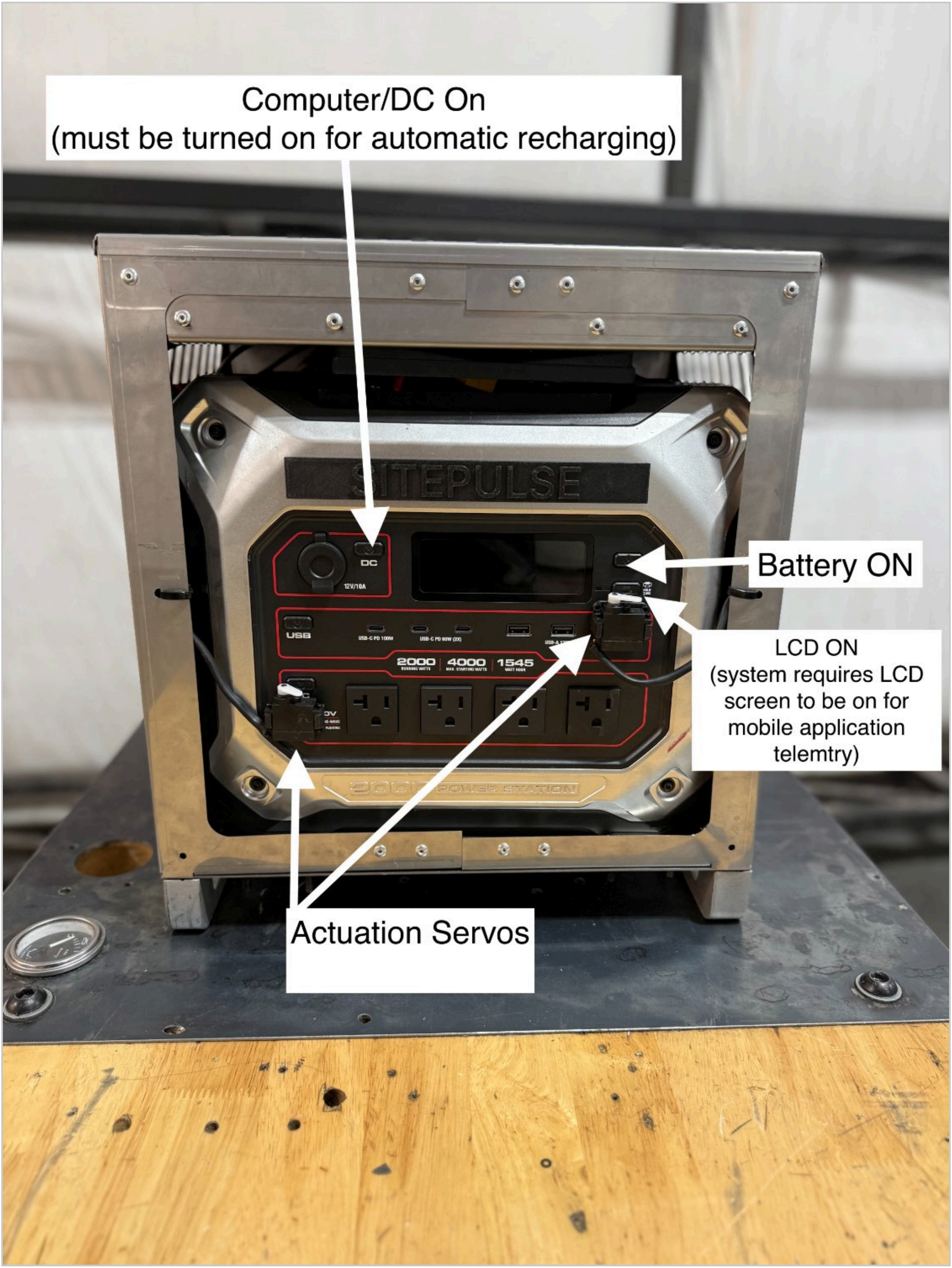

FIG. 02 — CONTROL PANEL. BOTH POWER BUTTONS, THE LCD, AND THE TWO SERVO ACTUATORS.

01 **Hold the main power button for about two seconds.** The battery comes online and the LCD wakes.

02 **Press the DC button.** This powers the onboard computer, which manages automatic recharging and all remote control.

03 **Confirm the LCD is lit.** The screen is the source of real-time battery data for the application. See section 04.

04 **Turn on the outputs you need.** AC, USB and 12 V each have their own button. Leave unused outputs off to preserve charge.

REQUIRED

Both the battery and the DC computer must be on for remote operation and automatic engine recharging. If only the battery is on, the unit is a manual power source and nothing will appear in the application.

03

Network and mobile application

The unit carries its own satellite link and broadcasts a local Wi-Fi network. Monitoring and control run through the web application from any browser.

ON-SITE WI-FI		APPLICATION	
NETWORK	SITEPULSE	ADDRESS	app.sitepulse.space
PASSWORD	12345678	SIGN IN	Use the account issued with your unit.

01 Join the SITEPULSE network when you are on site, or work over any internet connection when you are away — the unit reports through its own satellite link either way.

02 Open app.sitepulse.space and sign in. The dashboard shows state of charge, input and output power, engine status and fuel estimate.

03 If the unit shows offline, check sky visibility first, then confirm the DC computer button is on.

04

Telemetry and the LCD actuator

A servo actuator presses the LCD button on the battery. That press is what makes the battery management system publish accurate data. Everything downstream of it — state of charge, run time, fuel estimates — depends on the actuator working.

LCD ASLEEP	LCD AWAKE
The system falls back to a basic voltage calculation. Readings drift and will not match the true state of charge.	Coulomb-counted data from the battery management system. This is the accurate real-time number.

Reading state of charge

Telemetry vacillates depending on whether the LCD is active. Before you act on a number, press **Wake LCD** in the application and read the value that follows.

Testing the actuator

Test the button presser from the application before any extended remote deployment.

1. With the unit in front of you, trigger the LCD wake command from the app.
2. Watch the servo move and the LCD light up.
3. Confirm the state of charge in the app updates to the value on the screen.
4. If the servo does not move or the screen does not wake, do not leave the unit on an unattended deployment.

CRITICAL

If the actuator fails in the field, the unit keeps running but reports estimated data only. Treat all remote readings as approximate until the actuator is restored.

05

Charging inputs

Four ways to put energy into the battery. The engine runs automatically; the rest are connected by hand.

SOURCE	RATING	NOTES
Engine	Automatic	Starts and stops on its own when the DC computer is on.
Solar	800 W max	Connectors and leads are stowed under the engine. Do not exceed 50 VDC; when running more than one panel, use identical panels.
Wall	1500 W max	Standard 110/120 V outlet. Roughly 1.5 hours to full. Use the supplied cable only.
Vehicle	12 V only	Slow. Run the vehicle while charging. Never charge from a 24 V system.

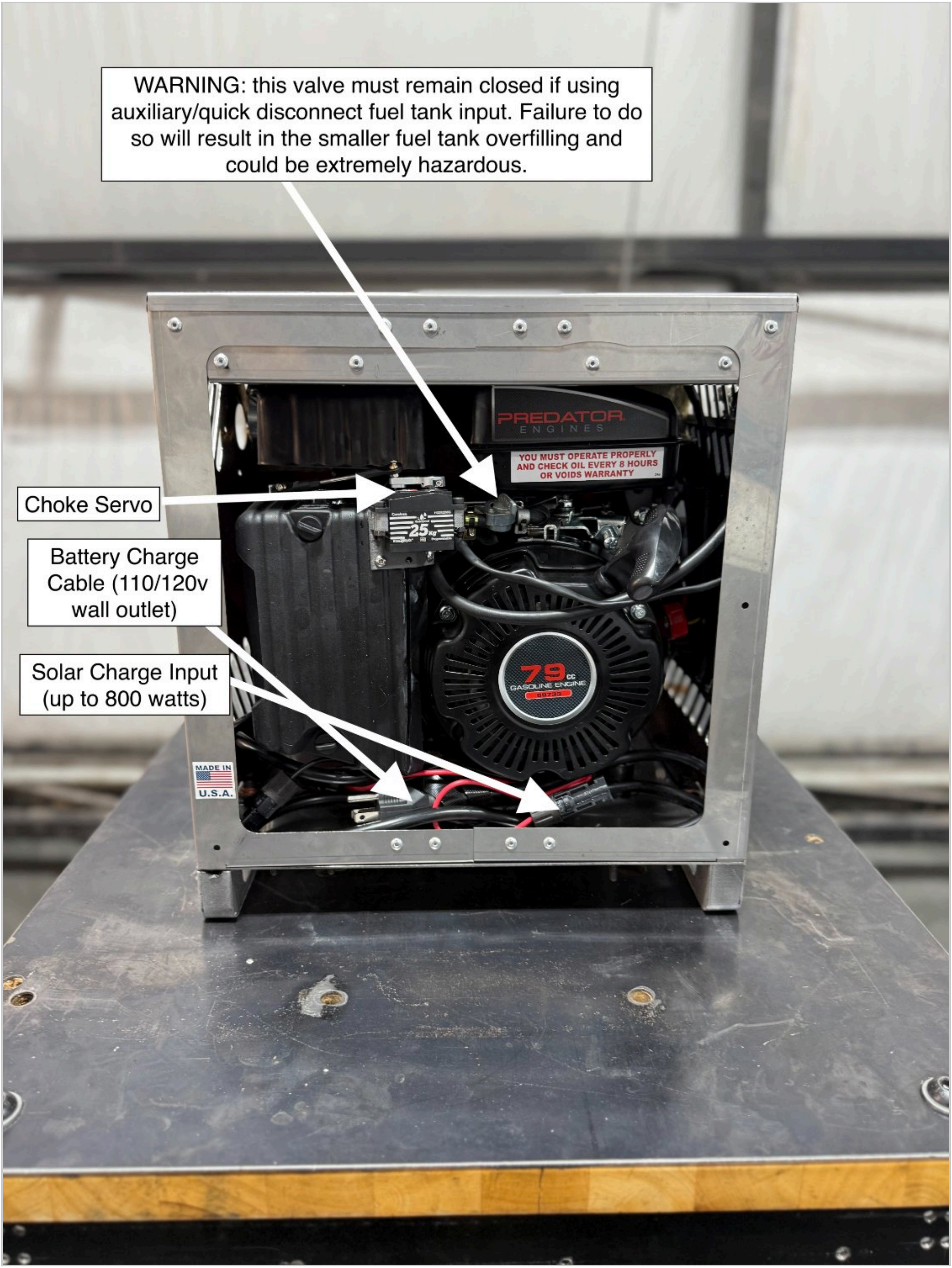

FIG. 03 — ENGINE BAY. CHOKE SERVO, WALL CHARGE CABLE AND SOLAR INPUT; CHARGE LEADS STOW BELOW THE ENGINE.

06

Fuel system and quick disconnect

The engine has a small onboard tank for short runs. Extended autonomous operation runs off an external 5-gallon metal jerry can through the fuel quick disconnect.

WARNING

The engine fuel valve must remain closed whenever the auxiliary tank is connected through the quick disconnect. Leaving it open will overfill the small onboard tank and is extremely hazardous.

FIG. 04 — FUEL QUICK DISCONNECT FOR THE AUXILIARY TANK INPUT.

Connecting the auxiliary tank

1. Stop the engine and let it cool. No smoking, no open flame.
2. Close the engine fuel valve.

3. Push the two halves of the metal fitting together until the collar clicks home. Tug the line to confirm it is locked.
4. Set the full 5-gallon can on level ground clear of the exhaust.
5. In the application, mark the tank as full. Run-time estimates are calculated from that reset against measured usage.

Disconnecting

Press down on the tab on the metal connector and slide it apart. The fitting seals both sides of the line automatically, so no fuel spills.

1. Stop the engine and let it cool.
2. Press the tab and slide the fitting out.
3. Cap the can and move it clear before restarting.
4. Open the engine fuel valve only if you intend to run on the onboard tank.

FUEL

Fresh 87+ octane unleaded gasoline, stabilizer-treated, with no more than 10% ethanol. Do not use E15, E20 or E85.

07

Engine service and oil intervals

The unit runs a Predator 79 cc horizontal engine, item 69733. Service follows the engine manufacturer's published schedule. Hours accumulate fast on autonomous deployments — track them, because the engine only runs when the battery calls for it.

INTERVAL	TASK
EVERY 8 HOURS	Check the oil level. The engine decal is explicit: operate properly and check oil every 8 hours or the warranty is void.
FIRST 20 HOURS	Break-in oil change. Drain warm and refill with SAE 10W-30.
100 HOURS / 6 MONTHS	Routine oil change, whichever comes first. Capacity is 0.37 qt (0.35 L).
EVERY 50 HOURS	Clean the air filter. Replace it if it is oil-soaked or torn. Dusty sites need it more often.
SEASONALLY	Inspect and re-gap the spark plug, check the fuel line and clamps, clear debris from the cooling fins.
STORAGE	Recharge the battery within 7 days of use, then every 3 to 6 months. Do not store below 32 °F.

COOLING

The cooling fan must start automatically whenever the engine runs. If it does not, shut the system down.

Changing the oil

1. Run the engine briefly so the oil is warm, then stop it and close the fuel valve.
2. Place a pan under the drain plug and remove the plug and the dipstick.
3. Drain fully, reinstall the plug, and refill with 0.37 qt of SAE 10W-30.
4. Check the level on the dipstick, run the engine, and check for leaks.
5. Log the hours and the date so the next interval is known.

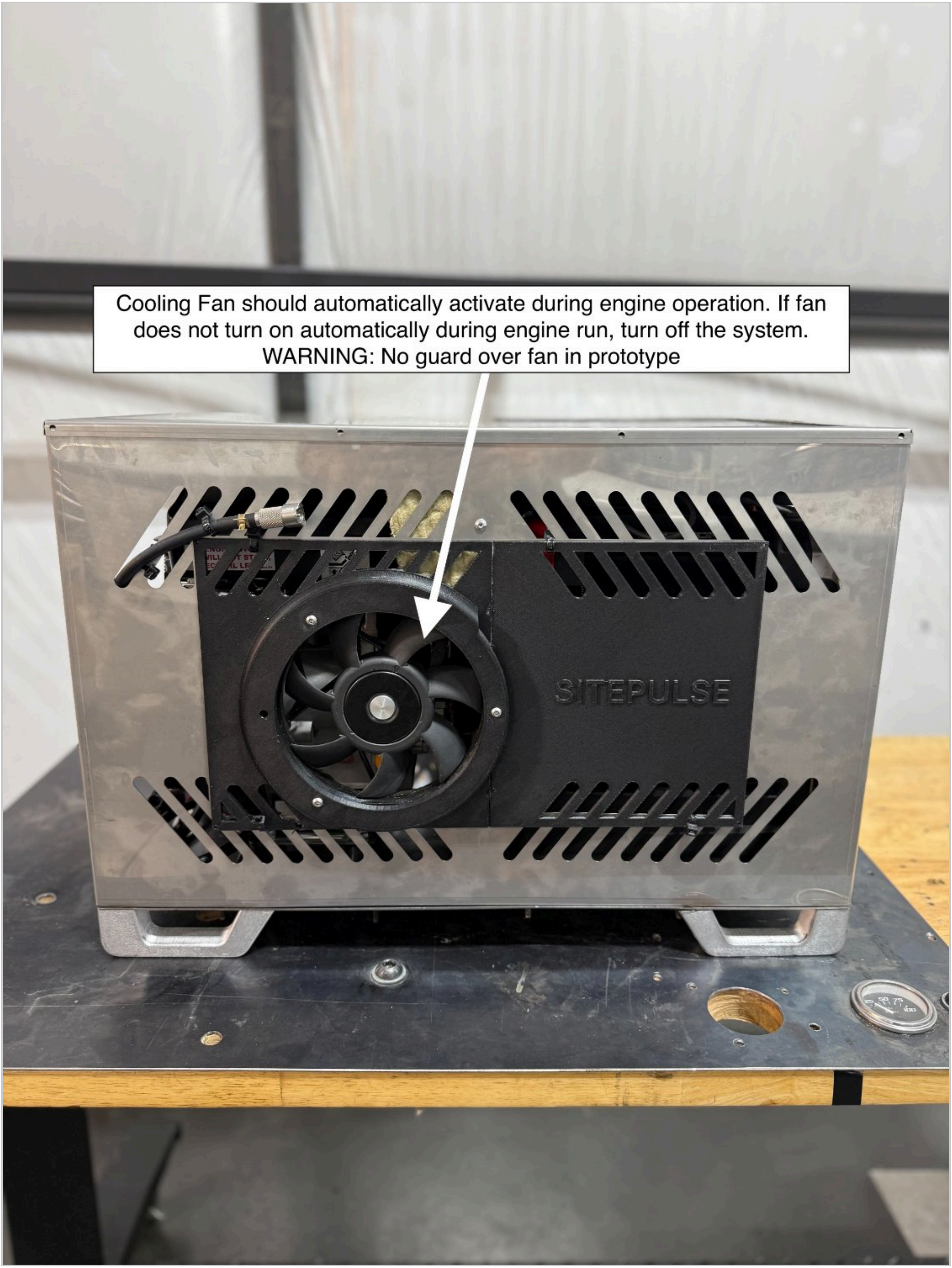

FIG. 05 — COOLING FAN ON THE ENCLOSURE SIDE.

08

Shutdown

1. Turn off the AC, USB and 12 V outputs.
2. Let the engine finish any charge cycle in progress, or stop it from the application.
3. Press the DC button to shut down the onboard computer.
4. Hold the main power button until the battery powers off.
5. If the unit is going into storage, disconnect the auxiliary tank and top the battery up first.

09

Safety

DANGER — CARBON MONOXIDE

Engine exhaust contains carbon monoxide, a poison you cannot see or smell. Using the engine indoors can kill in minutes. Never operate inside a home, garage, tent or trailer, even with doors and windows open. Run outside only, away from windows, doors and vents.

No fan guard on the prototype. Keep hands, tools and loose clothing away from the cooling fan whenever the system is powered.

Hot surfaces. The muffler and shroud stay hot well after shutdown. Let the engine cool before fueling or servicing.

Fuel handling. Refuel and connect tanks with the engine off and cool. No smoking, no sparks, no open flame. Wipe spills before restarting.

Do not open the battery housing. There are no user-serviceable parts inside and internal components can be energized even with the unit off.

Do not exceed the rated output. 2000 W continuous, 4000 W starting. Add loads one at a time and watch for an overload indication.

Not for building wiring. This is a portable unit and must not be connected into a building's electrical system.

Weather and temperature. Keep the battery dry. Do not charge below 32 °F or operate above 104 °F.

Inspect before each use. Loose hardware, damaged wiring, cracked parts, fuel weeping at a fitting — correct any of these before starting.

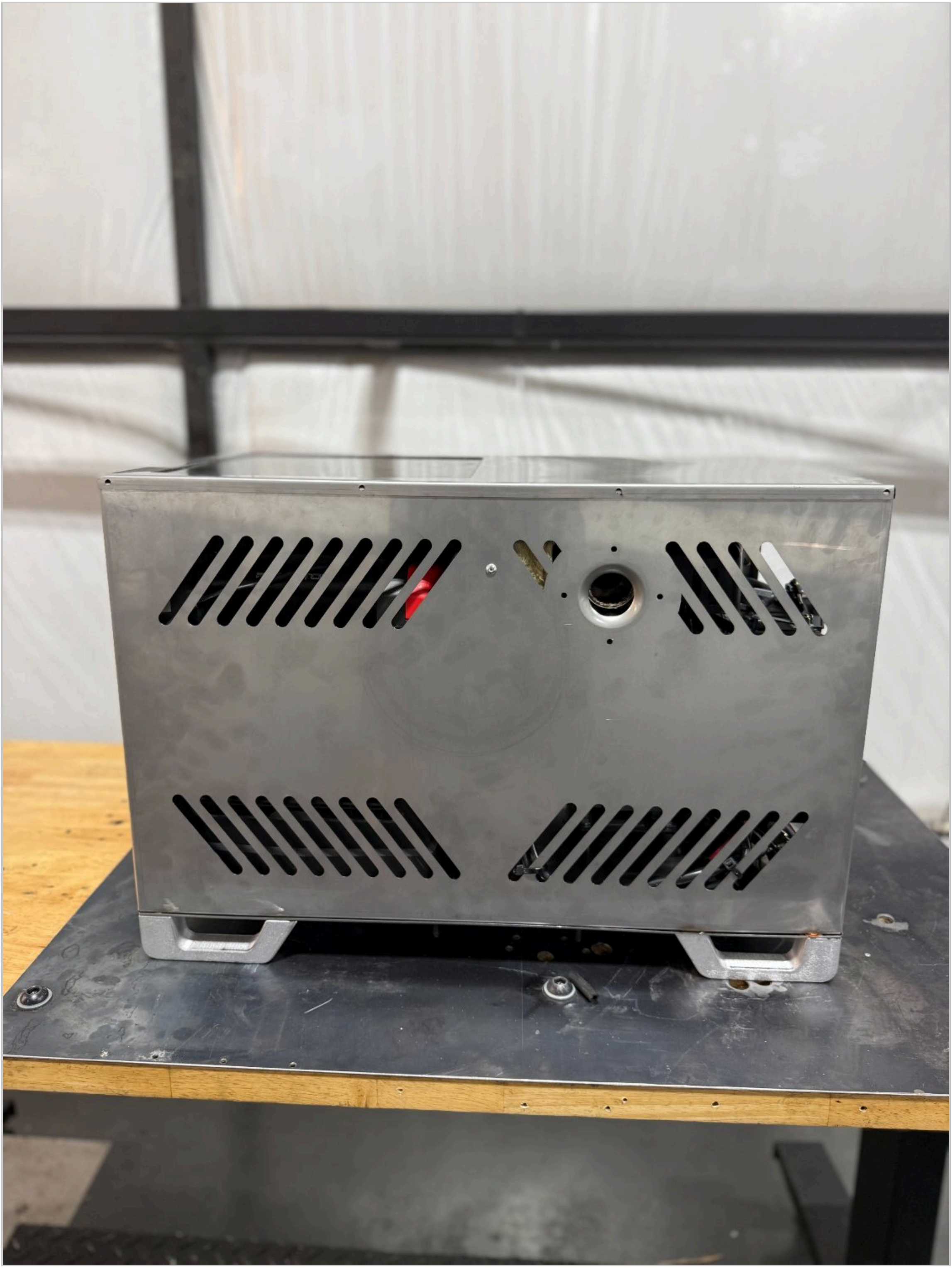

FIG. 06 — EXHAUST SIDE. KEEP THIS FACE CLEAR OF STRUCTURES, INTAKES AND OCCUPIED SPACE.

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Quick reference

START

1. Outside, clear view of sky
2. Hold main power ≈ 2 s
3. Press DC computer button
4. Confirm LCD lit
5. Enable outputs

BEFORE LEAVING SITE

1. Test LCD actuator from the app
2. Engine fuel valve closed
3. Aux tank connected and full
4. Mark tank full in the app
5. Confirm unit online

ACCESS

WI-FI	SITEPULSE
PASS	12345678
APP	app.sitepulse.space

SERVICE

OIL CHECK	every 8 h
OIL CHANGE	20 h, then 100 h / 6 mo
OIL SPEC	SAE 10W-30, 0.37 qt
FUEL	87+, ≤10% ethanol

If something looks wrong

SYMPTOM	CHECK
Unit offline in the app	Sky visibility, then the DC computer button.
State of charge jumping	Wake the LCD and re-read. The sleeping estimate is voltage-based.
Battery not recharging	DC computer on, fuel available, engine fuel valve set correctly for the tank in use.
Fan not running with the engine	Shut the system down. Do not run without cooling.
Engine will not start	Oil level and fuel first — the engine is protected against both.
Output cuts out under load	Total draw against 2000 W continuous. Remove a device and reset the output.